

IMG President's Innovation Fund

Year 1 Report

March 2026

Project Lead/PI: Shauna Heron, School of Social Sciences and Supervisor Dr. Meng Cheng Lau, School of Engineering and Computer Science

Project Title: Emotionally Adaptive Robotics to Increase Trust in Human-Robot Collaboration

Report prepared by Shauna Heron.

Section 1: Progress on Milestones and Deliverables

We are pleased to report that all project milestones were achieved, culminating in the production of a preprint manuscript published on arXiv, with journal submission imminent. This project represents a substantive contribution to the human-robot interaction (HRI) literature and has laid the groundwork for a planned PhD-level research program with direct relevance to safety-critical industrial environments, including the mining sector. The work also exemplifies the kind of cross-disciplinary collaboration — bridging applied psychology and robotics — that Laurentian University's strategic plan envisions, with research outputs that carry implications for both academia and industry.

Phase 1: Setup and System Development (May–November 2025)

The project launched as planned with task design, model research, and ethics submission completed in May–June 2025. Ethics approval was granted in July 2025 (Laurentian University Research Ethics Board, ROMEO File: #6021966). The Misty II robotic platform was acquired in July 2025; notably, the team learned that the unit price would increase substantially from approximately \$4,000 USD to \$7,995 USD effective August 1, 2025. Through rapid coordination among the research team and Laurentian's administration, the purchase was completed prior to the price increase, ensuring the project's primary equipment budget was used efficiently and that the approved project scope remained achievable.

Following hardware arrival, development of the Emotion-Adaptive AI system (EAI) continued through October and into early November 2025. The final system integrated the Misty II platform with an offboard dialogue management pipeline built in Python, incorporating the following components:

- Deepgram Nova-2 speech recognition API for speech-to-text (STT) processing;

- Google's Gemini language model (gemini-2.5-flash-lite) via the LangChain orchestration framework for dialogue management and robot facial expressions and head and arm movement;
- A DistilRoBERTa-based emotion classification model for real-time affect inference.

All interaction decisions, including spoken-language understanding, task state tracking, affect-responsive dialogue, and nonverbal behaviour generation, were executed autonomously without human intervention. The EAI system was integrated with a Flask-based web interface for experiment control and data logging, and was refined through a series of internal lab sessions that informed adjustments to dialogue design, interaction timing, and error handling strategies.

Phase 2: Data Collection (November–December 2025)

In-person research sessions were conducted at Laurentian University between November and December 2025. A total of 29 participants were recruited from the Laurentian University community via the SONA system and word of mouth. Following a pre-screening questionnaire administered through Qualtrics, participants completed an in-person, multi-stage collaborative task with the Misty II robot operating under one of two conditions: a responsive, affect-adaptive interaction policy, or a neutral, reactive interaction policy. Sessions were conducted in a private room on campus and lasted approximately 30 minutes. Participant compensation was provided in the form of Tim Hortons gift cards (\$15 per participant; total: \$450 CAD).

Although recruitment proved more challenging than initially anticipated with an achieved sample of $n = 29$ (smaller than the originally projected $n \approx 60$), as discussed in Section 2, this and other challenges directly informed a productive and scientifically defensible reframing of the study's scope.

Phase 3: Data Analysis (December 2025)

Quantitative and qualitative analysis was completed in December 2025. All analyses were conducted in R (version 4.5.1) using the Tidyverse suite, with mixed-effects models fitted via lme4/lmerTest and Bayesian hierarchical models fitted via brms. Dialogue transcripts were manually coded using a structured scheme developed specifically for this study. All summary statistics, figures, and model outputs were generated within a reproducible Quarto framework.

Results indicated that participants who interacted with the responsive robot reported significantly higher post-interaction trust than those in the control condition. Critically, this difference was not explained by differences in task performance, suggesting that trust was shaped by the quality of the interaction process itself rather than by instrumental utility alone.

Analysis further revealed that communication quality moderated these effects — a finding with important implications for deployment in linguistically diverse environments such as mining operations, and one that informed the strategic reframing of the study as a methodologically rigorous pilot (see section 2).

Phase 4: Dissemination (December 2025-Present)

An initial draft of the manuscript was completed in December 2025. The manuscript was finalized in January 2026 and posted as a preprint in late February 2026:

Heron, S., & Lau, M. C. (2026). *Trust in Autonomous Human–Robot Collaboration: Effects of Responsive Interaction Policies* (arXiv:2603.00154). arXiv.
<https://doi.org/10.48550/arXiv.2603.00154>

A research poster is completed and will be presented at Laurentian University Research Week (end March 2026). Submission of the manuscript to *Computers in Human Behavior: Artificial Humans* (Elsevier) is planned for the coming weeks. We hope to reserve the remaining funds to cover open-access publication fees upon acceptance.

Section 2: Were Milestones Achieved as Expected?

Project milestones were achieved largely as planned. Where deviations from the original proposal occurred, they were managed proactively and, in each case, produced a stronger and more defensible research output than the original confirmatory design would have permitted. The key deviations are addressed below.

Study Scope: Strategic Reframing as a Pilot Study

As noted above, the final participant sample ($n = 29$) was smaller than originally projected ($n \approx 60$). While practical recruitment challenges contributed to this shortfall, the decision to reframe the study as a methodologically rigorous pilot was driven primarily by a substantive finding that emerged during data collection that would have been difficult to overcome within our planned timeline.

The central finding motivating the reframing concerned language proficiency and speech recognition reliability. Despite pre-screening for English language fluency, actual spoken-language proficiency varied considerably across participants. In a subset of sessions, severe and persistent speech recognition failures attributable to this variability prevented the experimental manipulation from operating as intended, introducing a meaningful source of noise into the data. Rather than treating this as an inconvenient artefact to be absorbed by a larger sample, the team elected to examine and report it carefully — a decision that proved scientifically

productive. The relationship between communication quality and trust outcomes emerged as one of the study's most consequential findings, with direct implications for the design of speech-based HRI systems intended for deployment in linguistically diverse workplaces such as mining operations. Addressing this boundary condition systematically, through improved language screening, adaptive STT pipelines, or multilingual interaction design, is a clear priority for the future follow-up study.

In addition, our current implementation of EAI used a syntax-based affect inference model in place of a fuller multi-modal architecture described in the original proposal; incorporating facial expression recognition and prosodic analysis alongside syntactic inference remains a priority for the next phase of the research, where the longer timeline of a PhD-level program would make this feasible. Taken together, the pilot design produced rigorous, interpretable findings within the scope of the project, laid the empirical groundwork for a larger and better-powered confirmatory study, and generated a published preprint that represents a substantive contribution to the HRI literature in its own right — strengthening the case for continued investment in this research topic.

Dissemination: On Schedule

The originally proposed dissemination timeline targeted manuscript preparation by December 2025. In practice, the manuscript draft was completed in December 2025, finalized in January 2026, and posted as a preprint in February 2026 — consistent with the proposed timeline. Poster presentation at Laurentian University Research Week is underway (March 2026), and journal submission to *Computers in Human Behavior: Artificial Humans* is imminent.

Equipment Procurement: Proactive Cost Management

The unanticipated price increase for the Misty II platform represented a significant budgetary risk. The research team's ability to act quickly and complete the purchase prior to the August 1 price increase — which would have more than doubled the equipment cost — ensured that the project's primary equipment objectives remained achievable within the approved budget. The actual equipment expenditure of \$6,029.01 represents a saving relative to the projected cost of \$8,171.00, and the Misty II platform will continue to serve as a long-term research asset at Laurentian University, supporting future HRI and robotics research.

Budget: Under Budget with Funds Reserved

Total actual expenditures to date amount to \$6,479.01 against the \$8,000.00 awarded. Participant compensation (\$450.00) was an unbudgeted but ethically appropriate expense consistent with standard research practice for in-person studies of this kind. Conference travel (\$1,000.00 budgeted) was not utilized during this grant period, as suitable conference

opportunities aligned with the manuscript submission timeline have not yet arisen. We are hoping that remaining funds can be held in reserve to cover open-access publication fees upon acceptance of the manuscript.

IMG President's Innovation Fund
Year 1 Report - Budget
March 2026

Project Lead/PI: Shauna Heron, MSc Candidate, Applied Psychology, School of Social Sciences
and Supervisor Dr. Meng Cheng Lau, School of Engineering and Computer Science
Project Title: Emotionally Adaptive Robotics to Increase Trust in Human-Robot Collaboration

Cost Category			Year 1 – Projected*	Year 1 – Actual spend
Personnel (salaries and benefits)		Number		
	Master's student(s)			
	Doctoral student(s)			
	Undergraduate student(s)			
	Post-doctoral student(s)			
	Other (e.g., teaching release)			
Equipment				
	Equipment costs		8171.00	6029.01
	Computer costs			
	User fees			
Materials and Supplies				
	Lab supplies			
Participant Compensation (\$15 Tim Horton's cards)	Other costs		0	450.00
Travel				
	Conference travel		1000.00	0
	Collaboration/ field work			

Knowledge Dissemination				
	Workshops			
<i>Publication imminent</i>	Publication costs*		1000.00	0
	Other costs			
Other expenses (please specify)				
Total funding requested			10171.00	
Actual funding received			8000.00	6479.01

* Year 1 projected values should reflect those in the last approved version of the budget submitted for the project.